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AVARE BIOTECH

Powered by MAKSA Technology

Field Validation Report

MAKSA Technology — Frozen-Thawed Bovine Semen São Paulo, Brazil, March 2026

Independent Evaluator:

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Objective

To verify that the Avare Biotech app, powered by MAKSA (Mobile-Assisted Key Semen Analysis) technology, correctly measures semen concentration, total motility, and progressive motility in frozen-thawed bovine samples — compared to both CASA (SCA©) and manual microscope counting.

Materials & Methods

Five frozen semen straws from five different proven bulls were thawed following standard protocol (37°C water bath, 45 seconds). Each sample was evaluated simultaneously by three methods: (1) Avare Biotech app powered by MAKSA, (2) CASA — Sperm Class Analyzer (SCA©), and (3) manual microscope counting by an experienced evaluator. Three parameters were assessed: concentration (M/mL), total motility (%), and progressive motility (%).

Results

Bull / Straw	Parameter	MAKSA	CASA	Manual	Diff %	Validation
Bull 1 Ferreiro	Concentration (M/mL)	**28.40**	30.6	30	±10-15%	✓ **acceptable**

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	Total Motility (%)	**71**	72.1	70	±10-15%	✓ **acceptable**
	Progressive Motility (%)	**33**	18.2	30	±10-15%	✓ **acceptable**
Bull 2 Milei	Concentration (M/mL)	**28.4**	31.2	29	±10-15%	✓ **acceptable**
	Total Motility (%)	**67**	73.8	65	±10-15%	✓ **acceptable**
	Progressive Motility (%)	**32**	27.6	40	±10-15%	✓ **acceptable**
Bull 3 Gun Powder	Concentration (M/mL)	**30.40**	36.7	33	±10-15%	✓ **acceptable**
	Total Motility (%)	**62**	68.9	60	±10-15%	✓ **acceptable**
	Progressive Motility (%)	**38**	32	30	±10-15%	✓ **acceptable**
Bull 4 Minotauro	Concentration (M/mL)	**26.4**	32.1	25	±10-15%	✓ **acceptable**
	Total Motility (%)	**75**	73.7	80	±10-15%	✓ **acceptable**
	Progressive Motility (%)	**25**	40.7	45	±10-15%	✓ **acceptable**
Bull 5 USP	Concentration (M/mL)	**30.40**	29.9	32	±10-15%	✓ **acceptable**
	Total Motility (%)	**77**	88.3	60	±10-15%	✓ **acceptable**
	Progressive Motility (%)	**33**	37.9	40	±10-15%	✓ **acceptable**

Key Findings

- All three parameters (concentration, total motility, progressive motility) showed acceptable correlation across all five samples when compared to both CASA and manual methods.
- The variation between MAKSA and reference methods remained consistently within ±10-15% — the industry-accepted threshold for field-grade semen analysis.
- MAKSA demonstrated practical field utility: no specialized slides, no expensive CASA equipment — only a smartphone and a standard microscope.

Conclusion

The MAKSA application provides reliable and consistent information about frozen semen quality when compared with both computerized analysis (CASA) and subjective visual

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assessment. All 15 measurements across 5 bulls fell within the ±10-15% acceptable variation range, confirming the technology's suitability for routine field-based semen quality control.

Why This Matters

Brazil performs approximately 80 million artificial inseminations per year. The vast majority of post-thaw semen evaluations are still done visually — subjectively and without documentation. MAKSA provides an objective, digital alternative that costs a fraction of traditional CASA systems, enabling any AI technician or field veterinarian to verify semen quality before every insemination.

Willian Vaniel Alves dos Reis, DVM, PhD *Professor of Animal Reproduction, Universidade Federal de Mato Grosso do Sul (UFMS), Brazil* Signature: _____ Assinado por: Willian Vaniel Alves dos Reis, DVM, PhD 1FE6AE1E988C475... March 14, 2026	**Maksim Mershiev** *CEO & Co-Founder, Avare Biotech* Signature: _____ Signed by: Maksim Mershiev 325996BF0B2B43A... March 14, 2026

This report was prepared based on independent evaluation. Results may vary with equipment, sample handling, and environmental conditions.

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